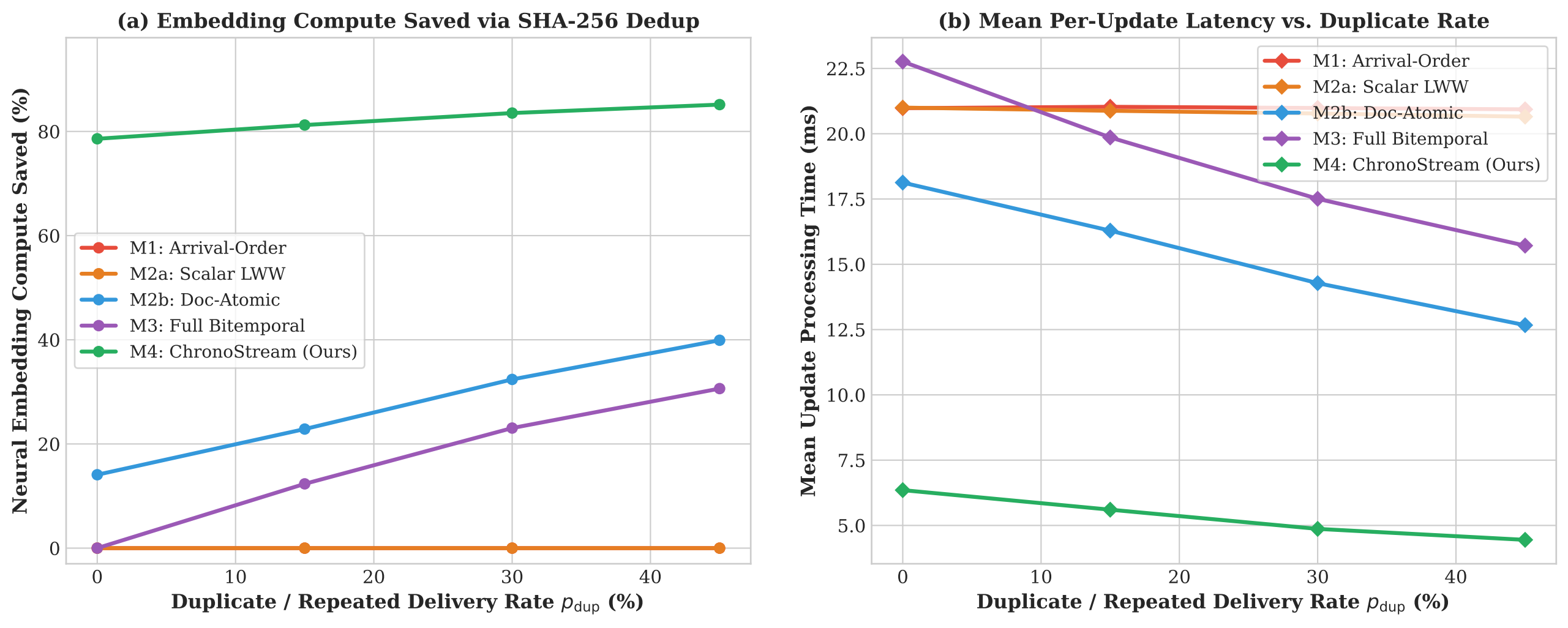
(a) Embedding Compute Saved via SHA-256 Dedup
Neural Embedding Compute Saved (%)
0
20
40
60
80
0
10
20
30
40
Duplicate / Repeated Delivery Rate p_{dup} (%)
M1: Arrival-Order
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemporal
M4: ChronoStream (Ours)
(b) Mean Per-Update Latency vs. Duplicate Rate
Mean Update Processing Time (ms)
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
0
10
20
30
40
Duplicate / Repeated Delivery Rate p_{dup} (%)
M1: Arrival-Order
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemporal
M4: ChronoStream (Ours)